Cells and Organisation Unit Overview KS3

About This Unit

This unit aims to give KS3 students an overview of the organisation of living things from single cells through to organ systems. They will start to look at how the structural differences between types of cells allows them to perform specific functions within the organism. They will explore how the skeletal and muscular systems work together to cause movement.

Notes

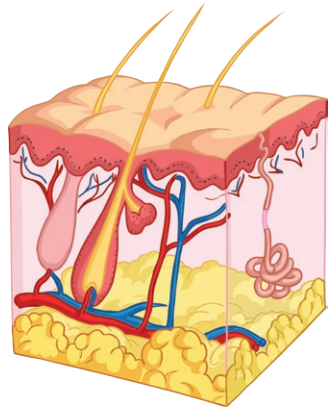
There is a Homework Grid that goes with this unit which lays out differentiated activities linked to each topic.

Lessons

1. How to Use a Microscope
2. Animal and Plant Cells
3. Bacterial Cells
4. Specialised Cells
5. Levels of Organisation
6. The Skeleton
7. Muscles

Core Practicals Included

Observing Plant and Animal Cells



Lesson 1: How to Use a Microscope

Included in this Pack

Teaching Ideas

Microscope Observation Activity Sheet

Parts of a Microscope Activity Sheet

Parts of a Microscope Picture Hotspot

How to Use a Microscope PowerPoint

Learning Objectives

To explore the importance of biological sciences and be able to use a microscope to magnify objects, to see in more detail.

Topics Covered

Using a light microscope.

Main Pupil Activities

Microscopes – Discuss the importance of the microscope and how it has led to scientific discoveries. Label parts of the microscope.

Using the Microscope to Observe Objects – Pupils look at objects and pre-prepared slides through their microscope and record what they observe on the activity sheet.

National Curriculum Aims

- Use a light microscope to observe and draw cells.
- Explain how to use a microscope to identify and compare different types of cells.

Resources

Lesson Pack

microscopes

pre-prepared slides

Objects to view through the microscope.

Health and Safety

Pre-prepared glass slides can cause cuts if broken. Model correct handling and storage of slides. Ensure floor is clear of obstacles. Remind students of fragility of glass slide. Dispose of any broken glass in glass bin and sweep surface clear.

Lesson 2: Animal and Plant Cells

Included in this Pack

Animal and Plant Cells PowerPoint

Teaching Ideas

Animal and Plant Cells Activity Sheet

Learning Objectives

To understand the differences between plant and animal cells.

Topics Covered

Structure of plant and animal cells.

Function of organelles in plant and animal cells.

Main Pupil Activities

Inside Cells – Students link descriptions of the main parts of the animal and plant cells to their location in the cell. They record the structure of both cells with a diagram and descriptions.

Journey into a Cell – Students follow the Journey into a Cell Activity Sheet to create their own guided tour around a plant or animal cell.

National Curriculum Aims

- Both plant and animal cells have a cell membrane, nucleus, cytoplasm and mitochondria. Plant cells also have a cell wall, chloroplasts and usually a permanent vacuole.

Resources

Lesson Pack

Lesson 3: Bacterial Cells

Included in this Pack

Bacterial Cells PowerPoint

Teaching Ideas

Bacterial Cell Diagram

Bacterial Cells Activity Sheet

Bacterial Cells Activity Sheet – Lower Ability

Bacterial Cells Extended Writing Mark Scheme

Learning Objectives

To explain how unicellular organisms are adapted to carry out functions that in multicellular organisms, are done by different types of cell.

Topics Covered

Structure and adaptations of bacterial cells.

Comparison of bacterial, animal and plant cell.

Main Pupil Activities

An Alien Cell – A chance to recap the structure of plant and animal cells.

I See, You Draw – A fun activity to introduce the structure of a bacterial cell, where one student describes the cell and the other must draw based on their description.

Extended writing – An activity to develop extended writing skills by asking students to compare plant, bacterial and animal cells.

Differentiation

A lower ability version of the Bacterial Cells Activity Sheet provides sentence starters to help students structure their extended writing.

National Curriculum Aims

- Explain how unicellular organisms are adapted to carry out their functions that in a multicellular organism are done by different types of cell.

Resources

Lesson Pack

optional: mini-whiteboards and pens

Lesson 4: Specialised Cells

Included in this Pack

Specialised Cells PowerPoint

Teaching Ideas

Cell Parts Can You Guess Cards

Specialised Cell Adaptations Activity Sheet

Specialised Cell Adaptations Activity Sheet Answers

Specialised Cell Adaptations Information Sheet

Specialised Cells Homework Activity Sheet

Differentiation

Sentence starters for extended writing are available on the Specialised Cells PowerPoint if required.

Learning Objectives

To explain how cells are adapted for their functions.

National Curriculum Aims

- There are many types of cell. Each has a different structure or feature, so it can do a specific job.
- Suggest what kind of tissue or organism a cell is part of, based on its features.
- Deduce general patterns about how the structure of different cells is related to their function.

Topics Covered

Specialised cells structure and function.

Resources

Lesson Pack

optional: sticky tack

Main Pupil Activities

Specialised Cells Treasure Hunt – Students search for information on the specialised cells that are hidden around the room and match it to the correct cells on their Specialised Cell Adaptations Activity Sheet.

Extended Writing – Students turn their collected facts into some extended writing about a specialised plant and animal cell.

Design a Specialised Cell – Students design a cell to do a specialised role in the body and can follow this up with research for homework to see how close their designs are to the real thing.

Lesson 5: Levels of Organisation

Included in this Pack

Levels of Organisation PowerPoint

Teaching Ideas

Levels of Organisation Activity Sheet

Levels of Organisation Activity Sheet Lower Ability

Organ Picture Cards

Differentiation

There is a lower ability version of the Levels of Organisation Activity Sheet which provides cloze statements to support students when writing keyword definitions. The sheet also provides a way to represent a hierarchy.

Organ Picture Cards can be used to help students recognise new organs.

Learning Objectives

To explain the hierarchy of organisation in a multicellular organism.

National Curriculum Aims

- Multicellular organisms are composed of cells which are organised into tissues, organs and systems to carry out life processes.

Topics Covered

The function of six organ systems.

The hierarchy of organisation.

Resources

Lesson Pack

optional: sticky tack

Main Pupil Activities

A Closer Look at Organs – These slides take the students through the six organ systems on the KS3 syllabus and identify an organ, tissue and cell from that system. The new information is recorded on the Levels of Organisation Activity Sheet.

Where Do These Organs Belong? – Students match other organs to the correct organ systems.

Hierarchy of Organisation – Students are provided with differentiated support to produce a hierarchy of organisation for the keywords in this lesson.

Lesson 6: The Skeleton

Included in this Pack

Skeleton PowerPoint
Teaching Ideas
Organ System Function Cards
Skeleton Activity Sheet
Skeleton Relay Diagram

Learning Objectives

To explain the role of the skeleton in the human body.

Topics Covered

The role of the skeleton and joints.

Main Pupil Activities

Skeleton Relay – This is a fun activity designed to get students learning the names of key bones by working as a group to reproduce a diagram of the skeleton from memory.

Skeleton Function – The students are given cards with functions of various organ systems on. They match these to the correct organ system to help them identify the roles that the skeleton plays and recap systems from the previous lesson.

Joints – Students use modelling clay to make simple versions of ball and socket and hinge joints.

Differentiation

There is a lower ability version of the Skeleton Activity Sheet which has keyword prompts to support students with writing scientific descriptions.

National Curriculum Aims

- The parts of the human skeleton work as a system for support, protection, movement and the production of new blood cells.
- Explain how a physical property of part of the skeleton relates to its function.

Resources

optional:
modelling clay
cocktail sticks
sticky notes

Lesson 7: Muscles

Included in this Pack

Muscles PowerPoint
Muscles Video
Teaching Ideas
Dissection Instruction Sheet
Muscles Activity Sheet

Learning Objectives

To explain how muscles interact with different tissues in the body to cause movement.

Topics Covered

Antagonistic muscle pairs.

Main Pupil Activities

Dissection – Students are guided through a chicken wing dissection that allows them to identify the main structures and model how antagonistic muscles work in pairs.

Muscles Activity Sheet – Students describe the look and feel of the tissues and record the role that each plays. They then explain why muscles need to work in pairs.

Differentiation

The lower ability version of the Muscles Activity Sheet lists the functions of each tissue type for students to match to the correct tissue and provides a cloze statement to help them explain how antagonistic muscle pairs work.

National Curriculum Aims

- Antagonistic pairs of muscles create movement when one contracts, and the other relaxes.
- Explain why some muscles contain muscle tissue.
- Explain how antagonistic muscles produce movement around a joint.
- Use a diagram to predict the result of a muscle contraction or relaxation.

Resources

Lesson Pack
chicken wings
dissection scissors
dissecting board or tray
bin bags
disinfectant in bucket/tub
disinfectant hand wash
optional:
gloves
aprons

Health and Safety

Please check the rules regarding carrying out this kind of experiment in your own learning environment. There is a risk that raw chicken is carrying salmonella, so do ensure that students are carrying out the dissection away from their work and personal belongings. Students should be reminded not to touch their face during the practical and disinfectant hand wash should be provided for hand washing before students return to their personal equipment. Benches should be wiped down with disinfectant after use.

Dissection tools could cause cuts. Model correct use and storage of dissection tools. Show students how to cut away from their body and other people. Ensure the floor is free of obstacles.